Virtual Try On Model Evaluation

Prepared for Vizzle. Assessment deliverable. Compiled 3 September 2026.
Scope: model research, shortlist against the accuracy, speed and cost gates, a category by category test across ten clothing types, and a single recommended model.

Summary. Three try on models were tested, all self hosted on one NVIDIA RTX 4090: IDM-VTON (with and without a category aware optimization layer), CatVTON, and Leffa. Every model clears both gates by a wide margin, so the decision is accuracy. Across three test rounds on full length person photos with the client garments, prompts and reference looks, IDM-VTON with the optimization layer scores a mean accuracy of 4.18 out of 5 and wins or ties ten of the eleven tested categories. CatVTON scores 3.53 and Leffa scores 3.33. **Recommended model: IDM-VTON, self hosted, with the category aware optimization layer.** Total testing spend was 3.16 US dollars against a 10 US dollar cap.

Contents

1. Market research and shortlist

Target unit cost is Rupees 4 per generation, about 0.045 US dollars per image. Target speed is under 15 seconds per generation. Two families of options were reviewed: commercial hosted APIs, and open source models that can be self hosted on a rented GPU.

1.1 Commercial and hosted APIs

Model or API	USD per image	Speed	Resolution	Notes	In budget
FASHN v1.6	0.075, falling below 0.04 at volume	5 to 10 s	864 by 1296	Best text and print fidelity. Lower body results mixed.	No at low volume
Kling Kolors v1.5	0.07	seconds	variable	Holds pose, skin and body. Can flatten fabric detail.	No
Google Nano Banana Pro (Gemini Flash Image)	about 0.06 to 0.14	seconds	high	Prompt driven, natural composites, not deterministic.	No
FLUX Virtual Try On Pro (fal)	0.0375 per megapixel	5 to 10 s	variable	Styling prompts. Held dress colour well.	Borderline yes

Model or API	USD per image	Speed	Resolution	Notes	In budget
image-apps-v2 (fal)	0.04	about 10 s	up to 4K	Simple two image input, full length placement.	Yes
Flux 2 LoRA try on (fal)	0.021 per megapixel	about 10 s	variable	Adjustable transfer strength.	Yes
Leffa (fal)	0.10	about 8 s	variable	Commercial cleared. Explicit garment type. Free when self hosted.	No on fal, yes self hosted

1.2 Open source models, self hosted on RunPod

Self hosting on a rented RTX 4090 is the way to reach the Rupees 4 target. The GPU rents for roughly 0.40 to 0.74 US dollars per hour, which is a fraction of a cent per image at about 10 seconds per image.

Model	Repository	Self host USD per image	Speed	Notes
IDM-VTON	yisol/IDM-VTON, ECCV 2024	about 0.002	10 to 15 s	Strong on upper body and dresses. Weak on saree and kurti out of the box.
CatVTON	Zheng-Chong/CatVTON	about 0.001	5 to 8 s	Lightweight, fast, fewer parameters.
Leffa	franciszzj/Leffa, Meta	about 0.003	8 to 12 s	Upper, lower and dress modes. Good pose preservation.
OOTDiffusion	levihsu/OOTDiffusion	about 0.002	about 10 s	Half and full body modes. Held as a backup.
Kolors VTON weights	Kwai-Kolors	about 0.003	about 8 s	Open weights of the Kling model. Held as a backup.

1.3 Shortlist taken forward

IDM-VTON, CatVTON and Leffa were taken into full testing, all self hosted on one RTX 4090. This combination covers the strongest published accuracy for Indian wear and dresses (IDM-VTON), the fastest and cheapest option (CatVTON), and the best pose preservation (Leffa), while keeping every generation far inside both gates. FLUX Pro Virtual Try On and image-apps-v2 were researched and priced but not tested, because the fal account was unfunded and locked during the assessment.

1.4 Sources

- fal.ai, 10 Best Virtual Try On APIs 2026: https://fal.ai/learn/tools/best-virtual-try-on-apis-2026
- FASHN, Top 4 Open Source VITON Models: https://fashn.ai/blog/comparing-the-top-4-open-source-virtual-try-on-viton-models
- ionio.ai, VTON models compared: https://www.ionio.ai/blog/vton
- pixazo, Best Try On APIs 2026: https://www.pixazo.ai/blog/best-virtual-try-on-api
- fal model pages: fal-ai/leffa/virtual-tryon, fal-ai/kling/v1-5/kolors-virtual-try-on, fal-ai/fashn/tryon/v1.5

2. Test method and scoring rubric

Each shortlisted model was run over the ten clothing categories in the brief: saree, kurti, lehenga, top, jumpsuit, t-shirt, shirt, coat, jeans and trousers. Coat was additionally tested on a female model, listed as coat (women), giving eleven test cases per model. The same person photo and the same garment photo were used across models within each category, so the only variable is the model.

2.1 Person and garment photos

Both person photos are full length, from head to feet. The female model wears a beige kurta set. The male model wears a black polo shirt, cream chinos and white trainers on a plain background. The male base was changed to this casual full length photo during round 3, so that grey jeans and charcoal trousers show with proper contrast and the whole leg and the feet stay in frame. Menswear was run again on this base. Womenswear was left unchanged from round 2. Garment photos are one product shot per category. Client supplied reference images were used to define the target look for each category.

2.2 The optimization layer

For IDM-VTON, an optional optimization layer was built for the harder categories. It has no fine tuning and no training. It applies a category aware full body mask so long garments are not capped at the waist, the client written prompt for each garment compressed to the 77 token limit of the text encoder, tuned diffusion steps and guidance, a mask union that absorbs a hanging dupatta or scarf so it does not leave a floating fabric artefact, and a light face and hair paste back on the full body runs to hold identity. The effective IDM-VTON result uses the optimized output for saree, kurti, lehenga, jumpsuit, coat and coat (women), and the baseline output for top, t-shirt, shirt, jeans and trousers.

2.3 Scoring rubric

Each result was scored 1 to 5 on five axes, where 5 is best. The mean is the average of the five.

Axis	What it measures
Fit	Garment shape, length and placement on the body
Drape	How the fabric falls, pleats and folds
Texture fidelity	Colour, pattern and weave matching the garment photo
Artefact free	Absence of floating fabric, bleed, warping or extra limbs. 5 means none.
Face and body preservation	Identity, face and body of the person kept intact

2.4 Gates

Both gates are generation time under 15 seconds and cost under Rupees 4 per image. Every tested model on the RTX 4090 runs at 5.6 to 11.2 seconds of GPU time and Rupees 0.10 to 0.20 per image, which is an order of magnitude inside both gates. Because all models pass, the ranking is decided on accuracy alone.

3. Per category scores

Scores are from round 3. IDM-VTON rows use the optimized output where one exists. Womenswear rows for CatVTON and Leffa are round 2 outputs on the same unchanged female model.

Model	Category	Variant	Fit	Drape	Texture	Artefact free	Identity	Mean	Notes
IDM-VTON	saree	optimized	4	4	4	4	5	4.2	Floor length black saree with a matching black blouse and the pallu over the left shoulder. The mask fix keeps it clean. Face paste back preserves identity.
IDM-VTON	kurti	optimized	5	4	4	5	5	4.6	Knee length kurti kept over the churidar. Placket, collar and side slits present. Catalogue grade.
IDM-VTON	lehenga	optimized	4	4	4	4	5	4.2	Choli with a floor length embroidered flared skirt and a sheer dupatta.
IDM-VTON	top	baseline	4	4	4	4	5	4.2	Blue velvet wrap top with the side tie visible. Kept the palazzo and identity.
IDM-VTON	jumpsuit	optimized	4	4	4	4	5	4.2	Black sleeveless wide leg jumpsuit with a collar and a tie belt. Clean.
IDM-VTON	t-shirt	baseline	4	4	4	4	5	4.2	Clean black crew tee. The garment photo was black on black so it was rebuilt from a plain flat lay. Natural fit, kept chinos and trainers.
IDM-VTON	shirt	baseline	5	4	4	5	5	4.6	Blue and black plaid flannel button up over the chinos. Plaid crisp, full length, identity intact.
IDM-VTON	jeans	baseline	4	4	4	4	4	4.0	Denim applied with a visible weave and five pocket styling, full leg to the ankle. Renders a touch darker than the grey source.
IDM-VTON	trousers	baseline	4	4	4	4	4	4.0	Charcoal trousers applied cleanly, full leg, kept the polo and trainers.

Model	Category	Variant	Fit	Drape	Texture	Artefact free	Identity	Mean	Notes
IDM-VTON	coat (men)	optimized	3	3	4	4	3	3.4	Knee length beige overcoat worn open over the outfit. The casual base means no white shirt and tie shows through as in the reference. The half redrawn polo reads as a dark top tucked into dark trousers. Face also drifts slightly on the full body mask. Known residual: needs a formal base photo or a prompt directed model to reproduce the shirt and tie look.
IDM-VTON	coat (women)	optimized	4	4	4	5	5	4.4	Black belted knee length wool coat with a fur collar. Clean layered drape, identity intact.
CatVTON	saree	baseline	3	2	3	4	4	3.2	Black one shoulder gown with a thigh slit. Full length but no pleats, no pallu wrap and no blouse.
CatVTON	kurti	baseline	4	3	4	3	4	3.6	Full length kurti. The dupatta becomes a stiff cape on the arm.
CatVTON	lehenga	baseline	4	3	3	4	4	3.6	Full three piece. Dupatta cape artefact. Embroidery soft.
CatVTON	top	baseline	4	4	4	3	4	3.8	Velvet wrap top faithful. Dupatta cape artefact.
CatVTON	jumpsuit	baseline	3	3	4	3	4	3.4	Black jumpsuit. The overall mask leaves the beige palazzo bleeding through one leg.
CatVTON	t-shirt	baseline	2	3	3	3	4	3.0	Renders the black tee as a tan long sleeve mock neck. Colour and sleeve length both wrong. No text control to correct it.
CatVTON	shirt	baseline	4	4	4	4	5	4.2	Plaid flannel correct, chinos preserved, clean.
CatVTON	jeans	baseline	4	4	4	4	4	4.0	Grey denim applied with visible texture, full leg, polo kept.

Model	Category	Variant	Fit	Drape	Texture	Artefact free	Identity	Mean	Notes
CatVTON	trousers	baseline	4	4	4	4	4	4.0	Charcoal trousers applied cleanly, full leg.
CatVTON	coat (men)	baseline	3	2	3	3	4	3.0	With the upper body mask it becomes a short beige blazer with beige trousers, not a knee length coat. With the full mask it wipes the person entirely.
CatVTON	coat (women)	baseline	3	3	3	2	4	3.0	Short coat, dupatta drape artefact, the beige palazzo still showing.
Leffa	saree	baseline	2	2	3	3	4	2.8	Black asymmetric drape over the churidar, not a saree. Grey sheer dupatta artefact.
Leffa	kurti	baseline	3	3	3	2	4	3.0	Full length kurti with an oversized tan cape artefact from the dupatta.
Leffa	lehenga	baseline	4	4	3	4	4	3.8	Flowy floor length gold gown. Reasonable.
Leffa	top	baseline	3	4	4	3	4	3.6	Velvet wrap top with a large blue drape down the side.
Leffa	jumpsuit	baseline	3	3	4	2	4	3.2	Black jumpsuit with an orange fur blob from the dupatta at the shoulder.
Leffa	t-shirt	baseline	3	3	3	3	4	3.2	Barely changes the polo. Low garment fidelity.
Leffa	shirt	baseline	2	3	3	3	3	2.8	Over applies the plaid to the whole outfit, giving a plaid jacket and plaid trousers.
Leffa	jeans	baseline	4	4	4	4	4	4.0	Grey denim applied, full leg, polo kept. Improved on the full length base.
Leffa	trousers	baseline	4	4	3	4	4	3.8	Charcoal trousers applied, full leg. Improved on the full length base.
Leffa	coat (men)	baseline	3	3	4	4	4	3.6	Beige coat rendered closed as a coat suit, no shirt or collar showing.
Leffa	coat (women)	baseline	3	3	3	2	3	2.8	Black coat with a burgundy sheer dupatta artefact.

4. Model comparison

4.1 Speed and cost per generation

Measured on one RTX 4090 at 0.74 US dollars per hour. GPU seconds only. Rupees per image uses Rupees 65 per hour.

Model	Generation time	Cost per image
CatVTON	5.6 s	Rupees 0.10
Leffa, self hosted	6.6 s	Rupees 0.12
IDM-VTON baseline	6.8 s	Rupees 0.12
IDM-VTON optimized	7.7 to 8.6 s	Rupees 0.14 to 0.16

All values are far inside the Rupees 4 and 15 second gates. Even at a hosted serverless 4090 rate of about 1.10 US dollars per hour, the worst case is about Rupees 0.25 per image.

4.2 Mean accuracy rollup

Model	Categories	Mean accuracy	Weakest category
IDM-VTON, effective (optimized where available)	11	4.18	coat (men) at 3.4, no shirt and tie under the open coat
CatVTON, baseline	11	3.53	t-shirt at 3.0, colour and sleeve wrong, no text control
Leffa, baseline	11	3.33	saree at 2.8

For reference, the IDM-VTON baseline alone (no optimization) means 4.20 over the five categories where a baseline was scored, and the optimized set means 4.17 over its six categories. The effective figure of 4.18 is the one to compare against the other models, since it is the output a production setup would actually serve.

4.3 Round 3 dataset note

Coat is tested on both models. The male base changed to a casual full length photo, which fixed grey jeans and charcoal trousers but removed the shirt and tie that used to sit under the coat. IDM-VTON only repaints the coat region, so it cannot create a white shirt and tie that is not in the base photo. The opening of the coat therefore shows the half redrawn polo. This is an accepted limitation and is scored down. The women's coat does not have a shirt and tie in the brief and is unaffected.

5. Best model per category

Category	Winner	Score	Why
saree	IDM-VTON, optimized	4.2	Only result with a real floor length drape, a pallu and a matching blouse.

Category	Winner	Score	Why
kurti	IDM-VTON, optimized	4.6	Full length with a placket. The others add a dupatta cape.
lehenga	IDM-VTON, optimized	4.2	Choli, floor length embroidered skirt and a sheer dupatta. Cleanest.
top	IDM-VTON	4.2	Velvet wrap faithful, no side drape artefact.
jumpsuit	IDM-VTON, optimized	4.2	Clean wide leg jumpsuit. CatVTON bleeds the palazzo, Leffa adds a blob.
t-shirt	IDM-VTON	4.2	Clean black tee. CatVTON turns it tan long sleeve, Leffa barely applies it.
shirt	IDM-VTON	4.6	Plaid flannel crisp. CatVTON close at 4.2. Leffa turns it into a plaid suit.
jeans	IDM-VTON, CatVTON and Leffa tie	4.0	All three apply denim cleanly on the full length base.
trousers	IDM-VTON and CatVTON tie	4.0	Both clean. Leffa at 3.8.
coat (men)	Leffa 3.6, IDM-VTON optimized 3.4	3.4 to 3.6	Weakest category for every model. None reproduce the reference shirt and tie because the base is a casual polo. The Leffa closed coat scores marginally higher on the rubric. The IDM-VTON open knee length silhouette matches the client brief better. CatVTON is 3.0 with a short blazer.
coat (women)	IDM-VTON, optimized	4.4	Belted wool coat with a fur collar, clean. The others add a dupatta artefact.

IDM-VTON wins or ties ten of the eleven categories and is the clear best on the eight hardest. The one exception is coat on the male model, where no model does well.

6. Recommendation

6.1 Primary: IDM-VTON with the category aware optimization layer

Run IDM-VTON self hosted on one RTX 4090, with the optimization layer for the harder categories. The layer uses a category aware full body mask, the client prompts compressed to the 77 token limit of the text encoder, tuned steps and guidance, a mask union that removes hanging fabric artefacts, and a light face and hair paste back on the full body runs. There is no fine tuning and no training.

- Best or tied on ten of the eleven categories across three test rounds, and the clear best on the eight hardest.
- The only model that adds a matching blouse under the saree and renders the coat open.
- Self hosted cost is about 8 to 11 seconds per image and about Rupees 0.16 to 0.20 per image, an order of magnitude inside both gates, with no per call vendor fee and no rate limit.
- Accepted residual: coat on a male model. IDM-VTON repaints only the coat region, so with a casual base photo it cannot show a white shirt and tie through the open front, and the face drifts slightly on the full body mask. Two fixes, neither done in order to stay inside budget: use a formal male base photo wearing a shirt, a tie and trousers and run again coat only for about 0.30 US dollars, or use a prompt directed model such as FLUX Pro Virtual Try On, which needs fal funding. The women's coat is unaffected.

6.2 Fallback for high throughput: CatVTON

CatVTON is the fastest at about 5.8 seconds and the cheapest at about Rupees 0.10 per image, and it is solid on menswear and simple tops. It has no text control, so it is weak on Indian wear, the plain black tee and coats. Use it for a low latency or lowest cost tier where drape on Indian wear is not the priority.

6.3 Not recommended: Leffa

Leffa improved on jeans and trousers with the full length base, but it still over applies patterns so a plaid shirt becomes a full plaid suit, it barely applies plain tees, it renders the coat closed, and it adds hanging fabric artefacts. Its fal price of 0.10 US dollars per image is also over the Rupees 4 gate.

6.4 Not tested: FLUX Pro Virtual Try On and image-apps-v2

Both are hosted only on fal, and the fal account was unfunded and locked during the assessment. Researched cost is about Rupees 3.4 and Rupees 3.5 per image, just inside the gate. FLUX Pro is prompt directed like the IDM-VTON optimization and is the single biggest remaining quality lever for the coat worn open over a shirt and tie, for saree pallu realism, and for locking the face on full body runs. Funding fal with about 5 to 10 US dollars would allow this test.

7. Cost ledger

Hard cap 10.00 US dollars. Every RunPod invoice and any fal usage export is kept for reimbursement, which is payable only if the work is selected.

Date	Item	Detail	USD
2 September 2026	RunPod credit	Pre loaded, not a spend	plus 15.00
2 September 2026	RunPod RTX 4090 secure, EU-RO-1, plus 70 GB network volume	Four pods on one network volume across three test rounds: environment setup four times, smoke tests, IDM-VTON baseline and optimized with prompt and mask iteration, CatVTON, and Leffa in an isolated environment. Round 1 was ten categories, round 2 was eleven with full body models and coat on both, round 3 was a menswear run again on a casual full length male plus coat and prompt tuning. About 4.3 pod hours in total.	about 3.16
	fal.ai, FLUX Pro Virtual Try On and image-apps-v2	Account unfunded, zero balance and locked, so not tested	0.00
		Total testing spend	**about 3.16**

Verified against the RunPod balance: 15.00 US dollars falling to 11.84 US dollars, which is 3.16 US dollars of actual spend across three rounds, about 32 percent of the 10 US dollar cap.

7.1 Per generation unit cost, the figure the brief asks for

GPU seconds only, on one RTX 4090 at 0.74 US dollars per hour, which is about Rupees 65 per hour.

Model	Generation time	Cost per image
CatVTON	5.6 s	Rupees 0.10
Leffa, self hosted	6.6 s	Rupees 0.12
IDM-VTON baseline	6.8 s	Rupees 0.12
IDM-VTON optimized	7.7 to 8.6 s	Rupees 0.14 to 0.16

All are far inside the Rupees 4 and 15 second gates. Even at a hosted serverless 4090 rate of about 1.10 US dollars per hour the worst case is about Rupees 0.25 per image. The results log records the per run cost in US dollars and in Rupees, and the harness halts automatically if the cumulative spend ever reaches 10 US dollars.

7.2 Not tested, hosted only, fal account unfunded

Model	Researched price	Against the Rupees 4 gate
FLUX Pro Virtual Try On, fal-ai/flux-pro/v1/vto	0.0375 per megapixel, about Rupees 3.4	Just under
image-apps-v2, fal-ai/image-apps-v2/virtual-try-on	0.04, about Rupees 3.5	Just under
Leffa on fal, fal-ai/leffa/virtual-tryon	0.10, about Rupees 8.8	Over the gate, so self hosted instead

Compiled from the project files research.md, scores.csv, comparison.md and COST_LEDGER.md. Currency conversion uses 1 US dollar of about Rupees 88 and 1 Rupee of about 0.0113 US dollars.